



Name: _____

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This exam has 3 questions, worth 6 marks in total. You have 60 minutes. For multiple-choice questions, choose one answer.

- 1) A logistic classifier assigns the following disease probabilities to 20 validation patients. Each row groups patients with the same probability. [2 marks]

Probability p	Actual disease present	Actual disease absent
0.82	3	0
0.56	1	5
0.34	1	6
0.12	0	4

Predict disease present when $p \geq t$. Consider $t = 0.30, 0.50$, or 0.70 . The operating requirement is recall $\geq 80\%$. Among feasible thresholds, maximize precision; if precision is tied, choose the larger threshold.

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Accuracy} = \frac{TP + TN}{20}$$

An always-negative baseline predicts disease absent for everyone. Which policy follows the operating rule, and how does its accuracy compare with this baseline? Round percentages to one decimal place when needed.

- a) $t = 0.50$: recall 80%, precision 44.4%, accuracy 70%—below baseline.
- b) $t = 0.70$: recall 60%, precision 100%, accuracy 90%—above baseline.
- c) $t = 0.30$: recall 100%, precision 31.3%, accuracy 45%—below baseline.
- d) Keep the always-negative baseline: recall 0% and accuracy 75%.

- 2) A logistic model uses the engineered feature vector $\phi = (x_1, x_2, x_1x_2)$ and score $z = w_1x_1 + w_2x_2 + w_{12}x_1x_2 + b$. [2 marks]

Initially, $(w_1, w_2, w_{12}) = (0, 0, 0)$ and $b = \ln 3$. The initial probability is therefore exactly 0.75 for both training observations.

Observation	x_1	x_2	y
1	1	2	1
2	2	1	0

Perform one simultaneous batch gradient-descent update with learning rate $\alpha = 0.40$. For mean cross-entropy with no regularization, use

$$G_j = \left(\frac{1}{2}\right) \sum (p_i - y_i) \phi_{ij}$$

$$G_b = \left(\frac{1}{2}\right) \sum (p_i - y_i)$$

$$w'_j = w_j - \alpha G_j$$

$$b' = b - \alpha G_b$$

All gradients must use the old parameters. Then classify the new observation $(x_1, x_2) = (1, 1)$, predicting class 1 if its updated probability is at least 0.64.

Use $\ln 3 \approx 1.099$ and these sigmoid approximations:

$$g(-0.101) \approx 0.475; g(0.499) \approx 0.622$$

$$g(0.599) \approx 0.645; g(1.699) \approx 0.845$$

Which updated parameter tuple $(w'_1, w'_2, w'_{12}, b')$ and new prediction are consistent?

- a) (0.250, 0.050, 0.200, 1.199); new probability ≈ 0.845 , so predict 1.
- b) (-0.250, -0.050, -0.200, 0.999); new probability ≈ 0.622 , so predict 0.
- c) (-0.250, -0.050, -0.100, 0.999); new probability ≈ 0.645 , so predict 1.
- d) (-0.500, -0.100, -0.400, 0.899); new probability ≈ 0.475 , so predict 0.

- 3) Four logistic models were fitted using the same scaled features and training/validation split. Their objectives were *[2 marks]*

$$J = 0.1 \sum |w_j| + 0.2 \sum w_j^2$$

Validation loss is unpenalized mean cross-entropy; smaller is better. Listed zeros are exact.

Model	a	c	w ₁	w ₂	w ₃	w ₄	Validation loss
R	0	0.15	0.75	0.35	0.20	0.15	0.08
L	0.15	0	0.85	0	0	0	0.06
E1	0.10	0.15	0.60	0	0.20	0.10	0.12
E2	0.25	0.15	0.70	0	0.30	0	0.15

The selected model must use both L1 and L2 penalties with positive strengths, retain required features x_1 and x_3 with nonzero coefficients, and exclude at least one optional feature, x_2 or x_4 . Among models meeting every requirement, select the one with the lowest validation loss.

- a) E1 satisfies the requirements.
- b) E2 satisfies the requirements.
- c) L satisfies the requirements.
- d) R satisfies the requirements.